Ibrahim Isphahani

Mechanical & Robotics Engineer

235 W Van Buren St,
Chicago IL, 60607

Skills

TECHNICAL SKILLS

- CAD & Design: Siemens NX, SolidWorks, Engineering Drawings (Advanced)
- Programming: C++, MATLAB, Python (basic)
- Hardware & Robotics: Arduino, Microcontrollers, Sensor Integration, Robotics Systems, Raspberry Pi (Linux)
- Simulation & Analysis: ANSYS (FEA)
- Data & Tools: Microsoft Office, Teamcenter (PLM)

CORE STRENGTHS

- Problem Solving & Troubleshooting
- Strong Communication & Leadership
- Hands-on Engineering & Prototyping
- Project Coordination & Team Collaboration

Experience

Tractian — *Field Service Engineer*

June 2025 - January 2026

- Installed and commissioned vibration monitoring systems across large-scale industrial facilities, ensuring accurate sensor placement and system reliability
- Diagnosed mechanical anomalies using vibration data, identifying early-stage failures in rotating equipment and helping prevent unplanned downtime
- Configured and optimized cloud-based monitoring platforms, improving data visibility and system performance for client operations
- Collaborated directly with maintenance teams and plant managers to troubleshoot issues, deliver actionable insights, and improve equipment health
- Managed full deployment lifecycle including pre-install planning, on-site execution, and post-install system validation

International Inc — *Mechanical Design Engineer*

June 2023 - June 2025

- Designed and developed exterior aerodynamic roof-mounted components from concept through production, ensuring performance, durability, and manufacturability
- Created detailed 3D models and engineering drawings using Siemens NX, following GD&T standards and DFM best practices
- Managed sourcing and manufacturing processes, collaborating with vendors to optimize cost, quality, and lead times
- Led prototyping and validation efforts, identifying design improvements through testing and iterative development
- Troubleshot and resolved design and production issues, improving product reliability and performance
- Utilized Teamcenter for data management, version control, and cross-functional collaboration

Cobalt Robotics/Relay Robotics — *Senior Reserve Engineer/Reserve Engineer (Contracted)*

May 2022 -Jan 2025/April 2023 - March 2025

- Extensive hardware maintenance and assembly/disassembly of security robots on client sites.
- Maintained strong customer relationships and effectively translating their feedback to backend software engineers for continuous improvement.
- Diagnosing hardware and software issues on-site and implementing solutions based on instructions provided by backend engineers.
- Deploying robots to customer sites as well as shipping robots back to HQ
- Performed: Deployment, mapping, battery replacement, camera replacement, motor replacement, full sheet metal body replacement, overall maintenance and training of new engineers and more.

Molex — *Product Design Engineer Intern*

May 2022 - Aug 2022

- Designed and developed components for high-voltage busbar systems used in power distribution applications
- Created detailed 3D models and engineering drawings in Siemens NX, ensuring accuracy and manufacturability
- Supported prototyping and manufacturing of busbars, collaborating with engineering and production teams
- Applied engineering principles to optimize component performance, fit, and reliability in high-voltage environments
- Contributed to design iterations based on testing, feedback, and production constraints

Education

September 2018 - May 2023

University of Illinois in Chicago, Chicago IL- *B.S. Mechanical Engineering*

- Senior Design Project Manager - SAE Formula E Electronic Throttle Control
- SAE BAJA 2021 Team
- EDT Robotic Arm Team 2022
- Air Force ROTC Cadet: 2019 - 2021
- Army ROTC Cadet: 2021 - 2022

Other Experiences

- **Relay Robotics** — Field Reserve Engineer (2023–2025)
  Performed deployment, maintenance, and troubleshooting of autonomous delivery robots; promoted to trainer
- **Army National Guard** (PFC, E3) — 2021–2022
  Completed leadership and crisis management training in high-pressure environments
- **Invent Inc** — Stock Manager (2018–2022)
  Managed shipping and fulfillment operations; gained experience in eCommerce and Amazon logistics
- **SAE Formula E** — Senior Design Project Manager
  Led cross-functional engineering team; contributed to CAD, electrical systems, and hardware integration
- **Robotics Projects**
  Built Arduino-based automation systems with 3d printed parts and contributed to 6-axis robotic arm gearbox design
- **Navistar Robotics Competition**
  Designed and built VEX robot chassis; 1st place finals, 3rd place semifinals
- **Certifications**: Six Sigma Green Belt, GD&T